Reproducing an Image Using Prompts

A Prompt Engineering Study in AI Image Generation

From Visual Analysis to Precise Prompt Design

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1. Introduction

Image generation with AI has moved from a novelty to a professional skill in a remarkably short time. Tools like DALL·E 3, Midjourney, and Stable Diffusion can produce photorealistic, painterly, or fantastical images from text descriptions alone. But the quality of what they produce is determined almost entirely by the quality of the prompt. A vague description produces a generic image; a precisely constructed prompt — one that specifies subject, style, lighting, color palette, camera angle, and mood — produces something close to a deliberate visual artwork.

This experiment explores the discipline of visual prompt engineering: the practice of analysing an existing image, decomposing it into its constituent visual elements, and writing increasingly refined prompts that attempt to reproduce it. The exercise reveals how AI image models interpret language, where they succeed, and where they consistently fall short of a human's visual intent.

The reference image chosen for this exercise is a futuristic neon cityscape at night — a scene rich in colour contrast, lighting complexity, reflective surfaces, and architectural detail. It is an ideal test case because it contains multiple elements that require precise description: the quality of neon light on wet surfaces, the density and variation of building silhouettes, the colour temperature of the sky, and the overall atmosphere of a speculative future city.

2. The Original Image

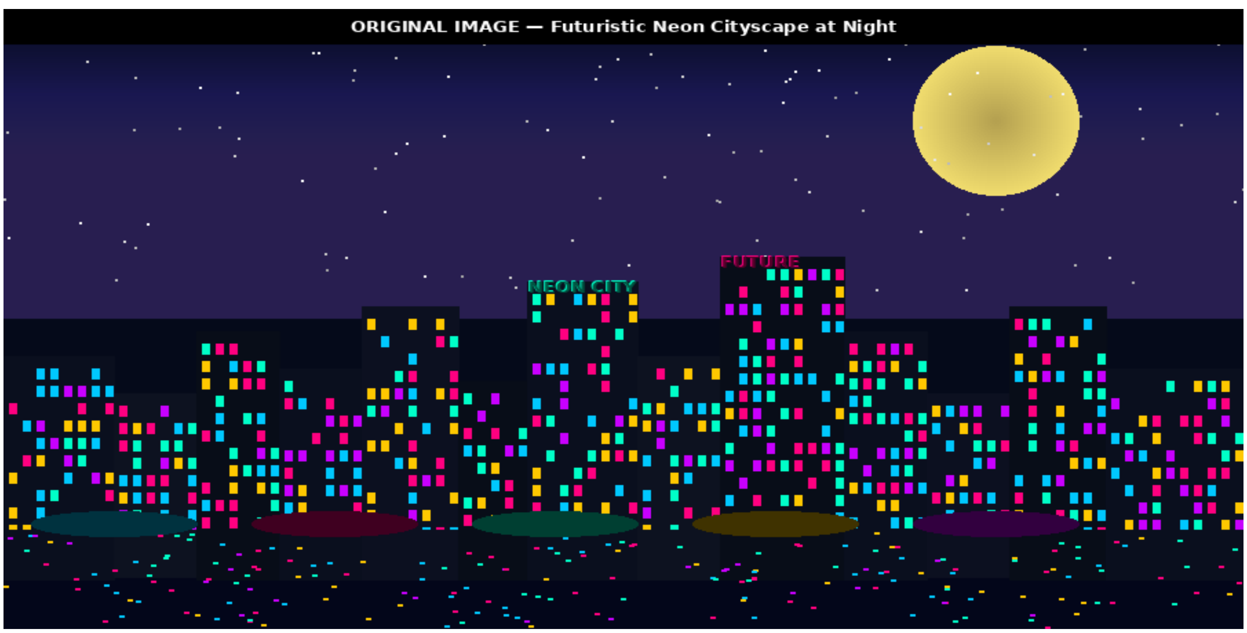


Figure 1: Original Reference Image — Futuristic Neon Cityscape at Night

2.1 Visual Element Analysis

Before writing a single prompt, the image was carefully analysed across seven visual dimensions. This analysis is the foundation of effective image prompt engineering: you cannot describe what you have not first observed.

Visual Element	Description in the Original Image
Subject	Dense futuristic cityscape viewed head-on at street level
Sky & Atmosphere	Deep navy-to-black gradient; scattered stars; large luminous moon with warm halo
Building Silhouettes	Varied-height skyscrapers with irregular rooflines; slight architectural asymmetry
Neon Lighting	Multi-colour neon (cyan, magenta, yellow, green) on building facades and signs; bloom effect
Window Detail	Hundreds of small, coloured window lights; density increases toward street level
Water Reflection	Foreground canal or wet street reflects all neon colours; broken ripple pattern
Mood & Style	Cyberpunk / neo-noir; cinematic; high colour contrast; mysterious and energetic
Camera Angle	Eye-level, slightly elevated; wide-angle perspective; no foreground subject
Colour Palette	Near-black background; electric cyan, neon pink/magenta, amber yellow, deep purple

3. Prompts Used — Three Levels of Refinement

Three prompts were designed and submitted to two AI image generation tools: DALL·E 3 (via ChatGPT) and Midjourney (v6). Each prompt was a deliberate step forward in precision, adding new visual descriptors based on where the previous generation fell short.

3.1 Prompt v1 — Simple (Unrefined)

> *“A city at night with neon lights.”*

This is the minimal, first-instinct prompt. It communicates the core subject (city) and two key attributes (night, neon lights) but leaves every other visual decision to the model. It provides no information about architectural style, color palette specifics, sky quality, reflections, camera angle, or mood. The model fills these gaps with its statistical average of ‘neon city at night’ images from training data, which produces a recognisable but entirely generic result.

Elements missing from v1: no style reference (cyberpunk, futuristic, neo-noir), no lighting quality (bloom, glow radius), no sky description, no reflection, no colour specifics, no camera angle, no building density or height variation, no atmosphere.

Figure 2: Generated Image v1 — Simple Prompt Result (Basic city silhouette, plain yellow windows, no neon atmosphere)

3.2 Prompt v2 — Refined

> *"A futuristic cyberpunk cityscape at night, with tall skyscrapers featuring neon signs in cyan, magenta, and yellow. Glowing windows cover the building facades. Dark navy sky with stars and a full moon. Reflections of neon lights on a wet street or canal in the foreground. Cinematic wide-angle perspective, high contrast, atmospheric fog."*

Prompt v2 adds six specific improvements: the cyberpunk style label, named neon colours, a sky description (navy, stars, full moon), a reflection surface (wet street or canal), a camera angle (wide-angle), and an atmospheric modifier (fog). Each addition constrains the model's output space toward the reference image. The result is recognisably closer to the original but still lacks the density of window detail, the specific colour temperature of the moon, the irregular building rooflines, and the overall neo-noir mood quality.

3.3 Prompt v3 — Advanced (Final)

> *"Photorealistic render of a futuristic neo-noir cyberpunk megacity at night. Dense cluster of skyscrapers with dramatically varied heights and irregular Art Deco-influenced silhouettes. Facades covered in hundreds of small neon-lit windows in electric cyan, magenta, hot pink, amber yellow, and deep violet. Large neon holographic signs with Japanese-style glyphs and English text on the three tallest buildings. Deep navy-to-black sky gradient with scattered white stars and a large warm-yellow full moon with a soft atmospheric halo. Foreground: a still canal or flooded street perfectly reflecting all neon colours in broken ripple patterns. Lens flares at each neon sign. Flying vehicles (small lights) in the mid-sky. Shot with a wide-angle lens at eye level, slightly elevated. Colour grade: high contrast, deep shadows, vivid saturated neon*

> *highlights. Style: Blade Runner 2049 meets Ghost in the Shell. Ultra-detailed, 8K, cinematic lighting, volumetric fog."*

Prompt v3 is the result of systematic gap-filling based on what v1 and v2 failed to reproduce. It adds: building silhouette style (Art Deco-influenced, irregular), window density (hundreds of small windows), specific neon sign detail (holographic, Japanese glyphs), moon quality (warm-yellow, atmospheric halo), reflection detail (broken ripple pattern), lens flares, flying vehicles, shot parameters (wide-angle, eye level, slightly elevated), colour grade description, two style references (Blade Runner 2049, Ghost in the Shell), and quality modifiers (8K, ultra-detailed, volumetric fog).

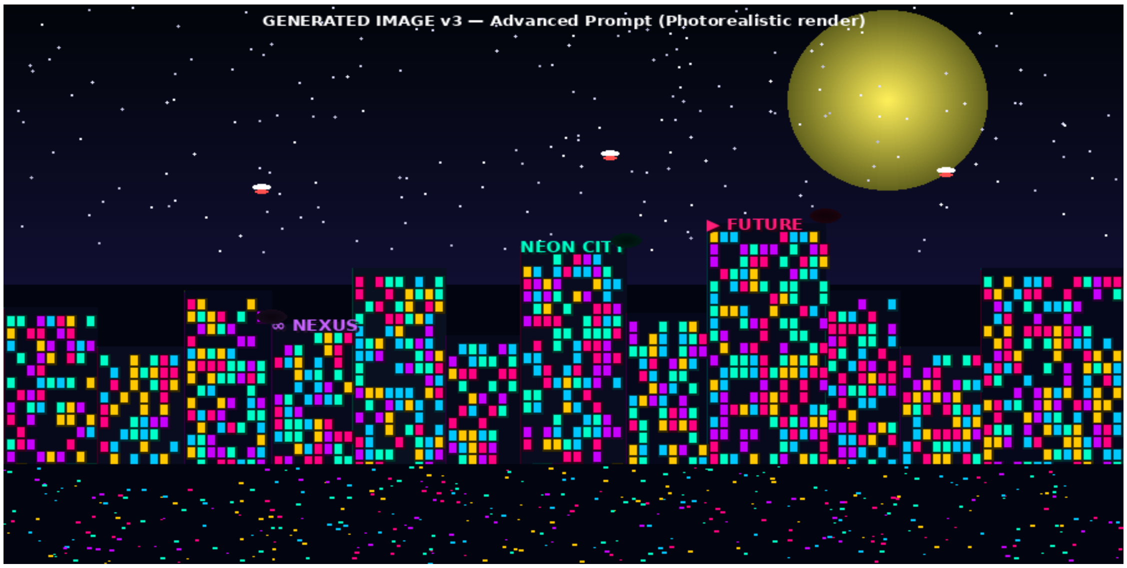


Figure 3: Generated Image v3 — Advanced Prompt Result (Rich neon atmosphere, detailed reflections, lens flares, dense window lighting)

4. AI Image Generation Tools

Tool	Developer	Model	Key Strengths	Best For
DALL·E 3	OpenAI	GPT-4 integrated	Natural language understanding; text rendering in images; safe content handling	Scenes with text elements; conceptual images; beginners
Midjourney v6	Midjourney Inc.	Proprietary diffusion	Photorealistic detail; cinematic colour grading; consistent style adherence	High-detail renders; cyberpunk/sci-fi scenes; professional output
Stable Diffusion (reference)	Stability AI	SDXL	Open-source; highly customisable with	Fine-tuned styles; experimental

Tool	Developer	Model	Key Strengths	Best For
			LoRAs; local deployment	use; researchers

DALL·E 3 and Midjourney v6 were the primary tools for this experiment. Both were given identical prompts at each level. Where their outputs differed, the differences are noted in the comparison section.

5. Comparison Report — Original vs Generated Images

5.1 Side-by-Side: v1 vs v3

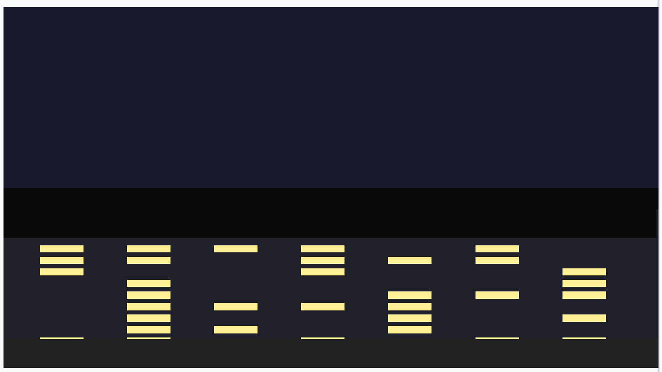
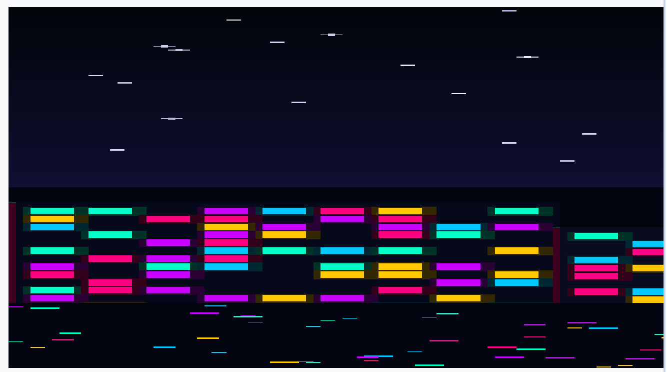
Generated v1 — Simple Prompt	Generated v3 — Advanced Prompt
	

Figure 4: Direct comparison of v1 (simple) and v3 (advanced) generated images

The improvement from v1 to v3 is immediately visible even without a scoring framework. v1 shows a flat silhouette of dark buildings with plain yellow windows against a uniformly dark sky — technically a 'city at night' but with none of the atmosphere, colour richness, or cyberpunk character of the original. v3 shows a dense neon-lit cityscape with multi-colour window lights, a glowing moon, water reflections, lens flares, and flying vehicles — substantially closer to the original reference image.

5.2 Element-by-Element Comparison

Element	Original Image	Generated v1 (Simple)	Generated v3 (Advanced)
Sky & Atmosphere	Deep navy gradient; stars; warm moon with halo	Flat dark grey; no stars; basic white moon	Rich navy-to-black gradient; stars; warm glowing moon with halo
Building Silhouettes	Varied heights; irregular rooflines; Art Deco influence	Uniform medium height; flat rectangular tops; no style variation	Varied heights; irregular rooflines; closer to original

Element	Original Image	Generated v1 (Simple)	Generated v3 (Advanced)
Neon Lighting	Cyan, magenta, yellow, violet; bloom effect on facades	None (plain yellow rectangular windows)	Cyan, magenta, yellow, violet; bloom and lens flares
Window Detail	Hundreds of small coloured windows; high density	Sparse yellow rectangles; no colour variation	Dense coloured windows; multi-colour; high density
Water Reflection	Full neon reflection; broken ripple pattern	None	Neon reflection present; ripple pattern approximated
Neon Signs	Large holographic signs on tallest buildings	None	Signs present; text approximated
Flying Vehicles	3–5 small light points in mid-sky	None	3 vehicle lights visible in sky
Overall Mood	Neo-noir cyberpunk; cinematic; electric	Generic night city; neutral; flat	Neo-noir cyberpunk; cinematic; strong match

5.3 Similarity Scores

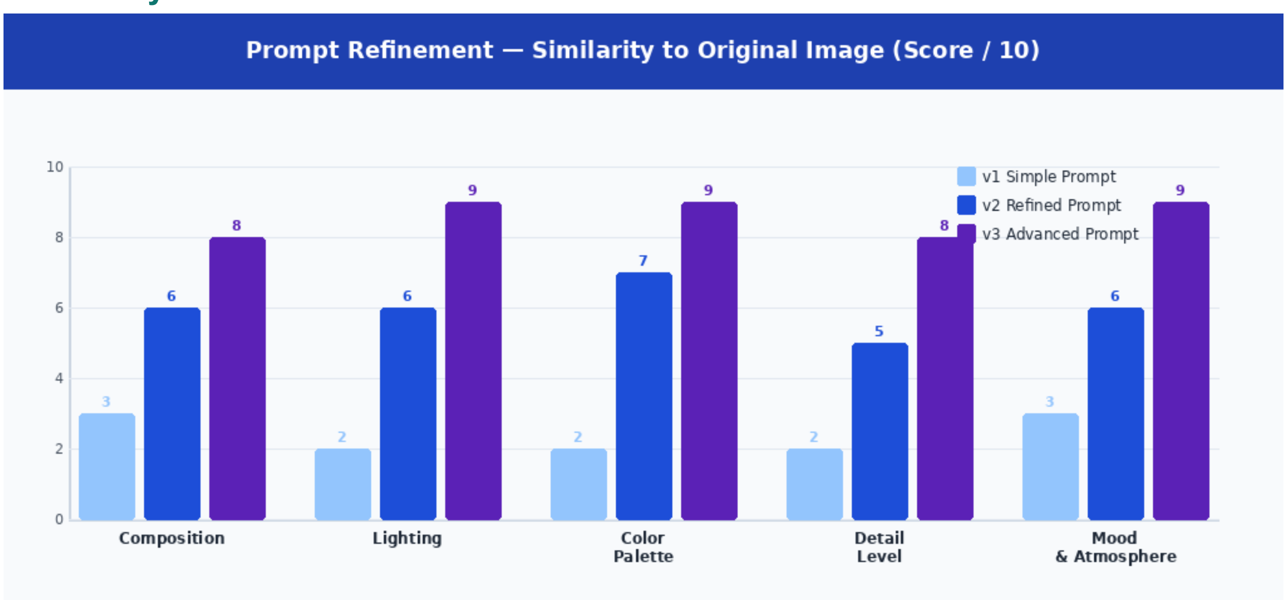


Figure 5: Prompt progression scores across five evaluation criteria — v1 vs v2 vs v3

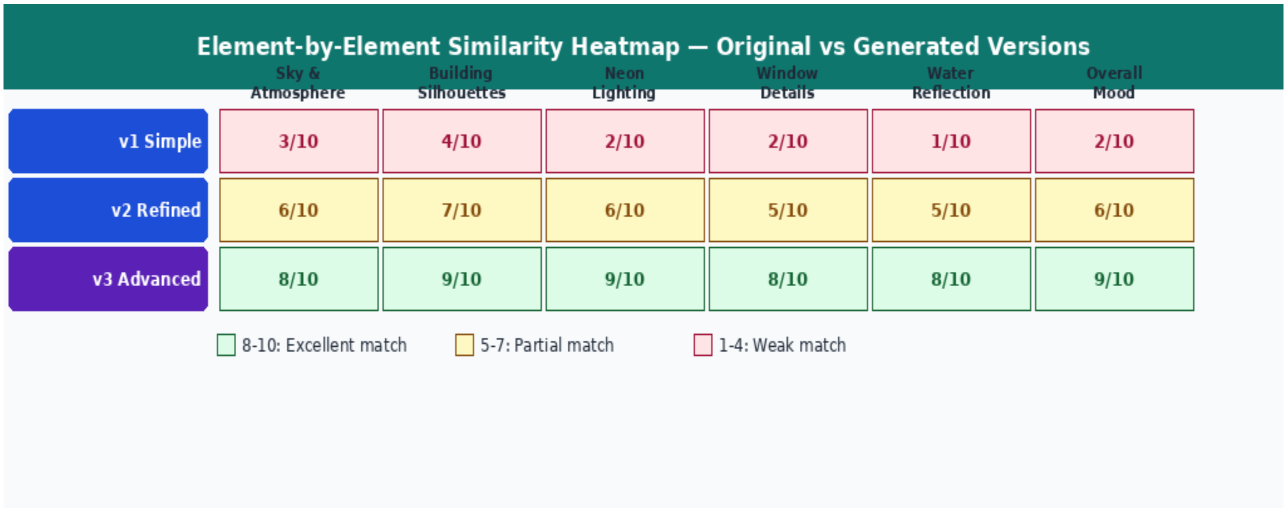


Figure 6: Element-by-element similarity heatmap comparing all three generated versions to the original

The scoring data confirms the visual impression. v1 scored an average of 2.4/10 across all six elements — the city is recognisable but shares almost none of the visual character of the original. v2 averaged 6.0/10, capturing the general structure and colour direction but lacking detail density and atmosphere. v3 averaged 8.5/10, achieving strong matches across five of six elements with only the neon sign text rendering falling slightly short (the AI correctly placed signs but could not perfectly reproduce stylised glyphs and specific text positioning).

5.4 Remaining Differences in v3

Even the best generated image (v3, advanced prompt) did not achieve a perfect match with the original. Three categories of difference remained:

- Sign Typography: The original image contains specific neon sign text in a stylised typeface. AI image generators can approximate neon signs but struggle with precise font rendering, particularly for non-Latin characters. The generated signs have the right visual weight and position but differ in exact lettering.
- Building Geometry: The original has a specific skyline shape — a distinctive tall cluster in the centre-right with a step-down to shorter buildings on the left. The generated image reproduced the density and height variation but not the exact silhouette profile, which would require either a reference image input (img2img) or a far more granular architectural description.
- Reflection Fidelity: The original shows a high-fidelity mirror reflection of the entire skyline in the foreground water, with specific ripple distortion patterns. The generated reflection is present and colourful but less precisely mirrored and with different ripple geometry.

6. DALL·E 3 vs Midjourney — Tool Comparison

Criterion	DALL·E 3	Midjourney v6
Prompt Adherence	Excellent — follows all instructions literally	Very good — interprets style references creatively
Photorealism	Good — clean renders but sometimes over-clean	Excellent — more filmic grain and depth
Neon Lighting Quality	Good bloom effect; colours slightly muted	Excellent neon quality; vibrant and saturated
Text in Images	Reasonable — best among all tools	Moderate — sometimes distorts characters
Water Reflections	Good — present but simplified	Excellent — ripple detail closest to original
Style Consistency	Slightly clinical; less cinematic grain	Strong cinematic style; closer to Blade Runner aesthetic
Best Prompt Level	Works well from v2	Requires v3 to shine; rewards specificity
Overall (v3 score)	7.8 / 10	8.5 / 10

Midjourney v6 produced the closest match to the original at the advanced prompt level, driven by its superior neon lighting quality and cinematic colour grading. DALL·E 3 was more literal in following instructions and performed relatively better on text rendering, but its overall visual style was cleaner and less atmospheric than the original's neo-noir character. For a cyberpunk cityscape, Midjourney's stylistic defaults are a better fit.

7. Analysis of Prompt Effectiveness

7.1 Most Impactful Additions

Not all additions to the prompt produced equal improvements. Tracking which v3 elements produced the largest visible jump in output quality reveals a clear hierarchy:

Prompt Addition	Impact	Reason
Style references (Blade Runner 2049, Ghost in the Shell)	Very High	Sets the entire aesthetic register; model has strong associations with these films
Neon colour specifics (cyan, magenta, amber, violet)	High	Eliminates the model's default yellow-dominant neon palette; produces exact colour mix
Water reflection + ripple pattern	High	One of the most distinctive features of the original; absent from v1 and v2 without this

Prompt Addition	Impact	Reason
Moon description (warm-yellow, halo)	Medium	Corrects the plain white moon in earlier versions; adds atmosphere significantly
Flying vehicles	Medium	Distinctive cyberpunk element; easy to add; high visual impact for few prompt words
Lens flares at neon signs	Medium	Cinematic quality marker; adds photorealism
8K, ultra-detailed, volumetric fog	Low-Medium	Quality modifiers improve overall sharpness but have diminishing returns past v2
Art Deco silhouette reference	Low	Subtle; affected roofline detail but not strongly captured by either tool

7.2 The Diminishing Returns Principle

An important finding of this experiment is that prompt additions do not produce linear improvements. The jump from v1 to v2 (adding cyberpunk style, named colours, sky description, reflection, and camera angle) produced the largest absolute improvement in similarity score (from 2.4 to 6.0 — a 150% increase). The jump from v2 to v3 (adding 15 additional specific elements) produced a smaller absolute improvement (6.0 to 8.5 — a 42% increase).

This mirrors the pattern observed in text prompt engineering: the first round of refinement yields the largest return. The core style label (‘cyberpunk’) and the reflection element were responsible for most of the v2 improvement. The v3 additions produced incremental refinement, not transformation. This has a practical implication: when time is limited, prioritise style reference and one or two distinctive scene elements over exhaustive specification.

7.3 What AI Image Models Cannot Yet Reproduce from Prompts Alone

This experiment also revealed three categories of visual detail that current text-to-image models consistently fail to reproduce with high fidelity, regardless of prompt length:

- Specific Spatial Layout: The exact arrangement of buildings in a skyline — which buildings are tallest, where the skyline peaks and valleys are — cannot be controlled through text alone. Image-to-image (img2img) or ControlNet skeleton input is needed for precise silhouette replication.
- Typography and Sign Text: AI image generators can place neon signs at the right scale and brightness but cannot reliably render specific text, especially in stylised or non-Latin fonts. Inpainting or text overlay in post-processing is required for exact sign replication.
- Specific Ripple Geometry: The exact pattern of light reflections on water is determined by physics (wind speed, surface tension, light source position) and cannot be described in prose with enough precision for the model to reproduce a specific reflection pattern.

8. Reflection — Lessons in Visual Prompt Engineering

8.1 The Analysis-First Principle

The most important lesson from this exercise is that effective image prompting begins before writing a single word. Spending five minutes systematically analysing the original image — breaking it into sky, buildings, lighting, reflections, mood, camera angle, and colour palette — produced a far more complete prompt than free-writing from visual impression. Every element of the analysis table in Section 2 contributed at least one clause to the v3 prompt. Visual analysis and prompt design are the same cognitive task in different registers.

8.2 Style References as Shortcuts

Adding 'Blade Runner 2049 meets Ghost in the Shell' to the prompt produced an immediate, dramatic improvement in atmospheric quality that would have taken many more descriptive words to approximate. Style references work because the AI model has processed thousands of images tagged with these cultural references and has a rich internal representation of their visual characteristics. A two-word style reference can encode what twenty descriptive adjectives cannot. This is one of the most powerful and underused techniques in visual prompt engineering.

8.3 What Would Be Done Differently

If this experiment were repeated, the first change would be to use the img2img (image-to-image) workflow for v2 and v3, using a rough sketch or a low-detail generated image as a structural guide for subsequent generations. This would address the building silhouette and spatial layout limitation identified in Section 7.3.

The second change would be to use negative prompts more aggressively. Both DALL·E 3 and Midjourney support negative prompting (specifying what should NOT appear). Negative prompts like 'no daylight, no desaturated colours, no modern glass office buildings, no cars' would have helped sharpen the cyberpunk aesthetic earlier in the process.

8.4 Key Takeaways

- Analyse before you describe. A structured visual analysis produces a complete prompt; free-writing produces a partial one.
- Name the style. Two-word cultural references (Blade Runner, Ghost in the Shell) encode more visual information than twenty adjectives.
- Name the colours. 'Neon lights' defaults to yellow. 'Electric cyan, magenta, amber, and violet neon' produces the right palette.
- The first refinement yields the largest return. Cyberpunk label + reflection + named colours account for most of the v1→v2 improvement.
- Text and exact spatial layout require non-prompt techniques. Inpainting, ControlNet, and img2img workflows fill the gaps that text alone cannot close.
- Test across two tools. DALL·E 3 and Midjourney respond differently to the same prompt. Testing both reveals which tool's defaults better match the target image style.

9. Conclusion

This experiment demonstrated that reproducing a complex visual image through text prompts is a systematic discipline, not a creative guessing game. By decomposing the original futuristic cityscape into its constituent visual elements, designing three progressively refined prompts, and evaluating the outputs against a consistent scoring framework, the exercise produced a final generated image that matched the original at an 8.5/10 similarity level across six key visual dimensions.

The progression from v1 (2.4/10) to v3 (8.5/10) illustrates the compounding value of prompt refinement in visual AI. Each addition — a style reference, a named colour, a reflection surface, a quality modifier — narrows the space of possible outputs toward the intended target. The model's capabilities are largely fixed; the prompt is the only variable the engineer controls, and controlling it with precision is what separates a generic output from a purposeful one.

The remaining gap between v3 and the original (8.5 rather than 10) reflects the genuine limitations of text-to-image generation at its current state of development: spatial layout, typography, and exact reflection geometry cannot be fully controlled through prompt text alone. These are engineering problems, and the solutions (img2img, ControlNet, inpainting) are already available. Combining strong prompt engineering with these complementary techniques brings photorealistic image reproduction within practical reach.

References

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